

Architecture

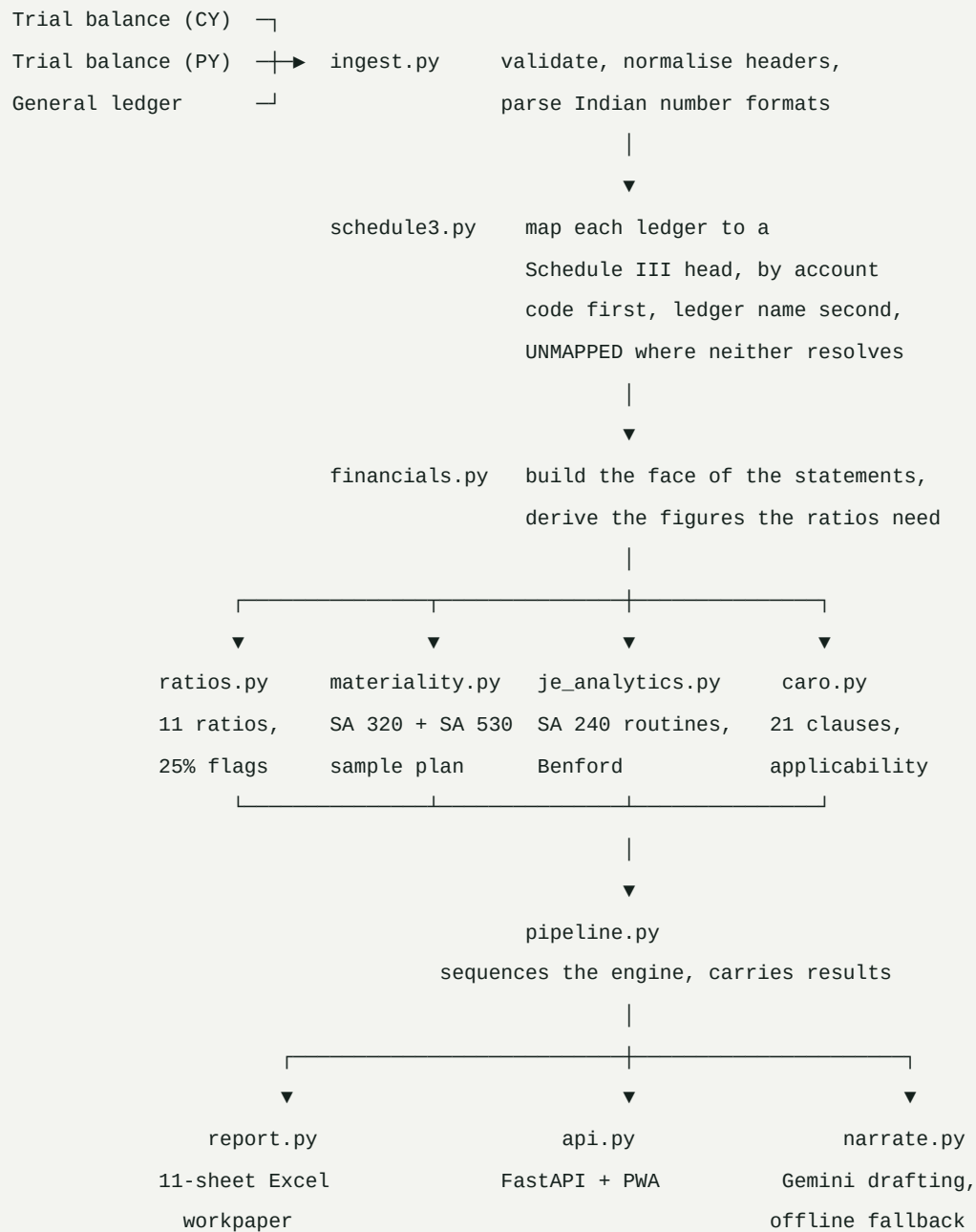
The governing rule

The engine computes. The model writes prose. Nothing else.

Every figure that reaches a workpaper, a ratio card or the face of the financial statements is produced by a pure Python function with a unit test. The language model is asked for exactly two things: an explanation in words, and correspondence. It is never asked to compute, to classify a ledger that reaches the statements, or to conclude on a CARO clause.

This is not a stylistic preference. A Chartered Accountant signs the report, and every figure in it must be traceable to a calculation the auditor can re-perform.

The pipeline



Why the modules split this way

Module	Holds	Deliberately does not hold
<code>ingest.py</code>	File reading, header aliasing, Indian number parsing, tally validation	Any judgement about what a ledger is
<code>schedule3.py</code>	The Schedule III head list and the mapping rules	Any arithmetic
<code>financials.py</code>	The face of the statements and the derived figures	Any ratio definition
<code>ratios.py</code>	The eleven ratios and the 25 per cent rule	Where the figures came from
<code>je_analytics.py</code>	The SA 240 routines and Benford	Any conclusion about an entry
<code>materiality.py</code>	SA 320 benchmarks and SA 530 sampling	The auditor's choice of benchmark
<code>caro.py</code>	The 21 clauses and applicability	Any answer to a clause
<code>formatting.py</code>	Indian digit grouping	Everything else
<code>narrate.py</code>	Prompts and drafting	Every figure it quotes comes from the engine
<code>pipeline.py</code>	Sequencing	No audit logic of its own

The consequence: `report.py`, `api.py` and `cli.py` all call `pipeline.py`, so the Excel workpaper, the web application and the command line cannot disagree with each other or with the tests.

Failure behaviour

Condition	What happens
Trial balance does not tally	Reported, with the difference. Never silently corrected.
A ledger cannot be mapped	Returned as <code>UNMAPPED</code> in a review queue. Never guessed.
Balance sheet does not tie	Reported, and reconciled against the unclassified value.
A ratio divides by zero	Returned as not computable, with the reason. Never zero.
The sample is impractically large	Computed, then flagged as an inefficient response with alternatives.
No API key configured	Drafting falls back to deterministic templates. The review still runs.
The drafting call fails	Caught. The analytical review is never taken down by a drafting failure.

Offline and on-premise

The whole analytical review runs with no network access. Drafting is the only component that reaches outside, and it degrades to templates when it cannot. For engagements where confidentiality terms rule out a third-party API, run a local model in LM Studio and point `AUDITLENS_MODEL` at it — the analytical result is identical either way, because the model never touches a figure.